

SALES PERFORMANCE ANALYSIS REPORT

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Internship Domain: Data Analytics

Project Title: Sales Performance Analysis using Superstore Sales Dataset

1. Executive Summary

This project analyzes the sales performance of a retail superstore using historical sales data from Kaggle. The objective is to understand business performance across different regions, product categories, customer segments, and time periods. The analysis identifies revenue trends, profitable business areas, low-performing products, and customer purchasing behavior.

The project includes data cleaning, feature engineering, KPI calculation, exploratory data analysis, and business visualizations to generate actionable business insights. The findings can help organizations like Alfido Tech improve sales strategies, optimize inventory management, and make data-driven business decisions.

2. Introduction

Sales performance analysis is an essential part of business intelligence because it helps organizations evaluate operational efficiency, identify growth opportunities, and improve profitability. By analyzing historical sales data, businesses can discover customer preferences, seasonal demand patterns, and regional performance differences.

This project performs an end-to-end analysis of a Superstore Sales dataset to identify factors influencing revenue and profitability.

3. Project Objectives

The primary objectives of this project are:

- Clean and preprocess the sales dataset.
- Analyze sales trends over time.
- Measure important business KPIs.
- Identify top-performing and low-performing products.

- Compare sales across regions and categories.
- Provide business recommendations to improve sales performance.

4. Dataset Description

Dataset Name: Superstore Sales Dataset

Source: Kaggle

Dataset size:

-Rows : 9800

-Columns: 18

Dataset

Link:

<https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>

The dataset contains transactional sales records including customer information, product details, shipping information, sales amount, profit, quantity sold, and order dates.

Major Features

Column	Description
Order ID	Unique order identifier
Order Date	Date of purchase
Ship Date	Date of shipment
Customer ID	Customer identifier
Customer Name	Customer information
Segment	Customer segment
Region	Sales region
Category	Product category
Sub-Category	Product sub-category

Column	Description
Product Name	Product description
Sales	Revenue generated
Quantity	Number of items sold
Discount	Discount offered
Profit	Profit earned

5. Data Cleaning & Preprocessing

Several preprocessing techniques were performed before analysis.

Cleaning Steps

- Removed duplicate records
- Checked missing values
- Converted date columns into datetime format
- Verified data types
- Created new time-based features:
 - Year
 - Month
 - Month Name
 - Quarter
- Calculated Shipping Time
- Prepared dataset for business analysis

These preprocessing steps improved data quality and ensured accurate analytical results.

6. Feature Engineering

Additional variables were created to enhance business analysis.

Created Features

- Order Year
- Order Month
- Month Name
- Quarter
- Shipping Time (Days)

These engineered features made it easier to identify seasonality, shipping efficiency, and yearly sales trends.

7. Key Performance Indicators (KPIs)

The notebook calculates several important business metrics.

KPI 1 – Total Revenue

Represents the total sales generated by the company.

Business Insight

A higher revenue indicates strong business performance and customer demand.

KPI 2 – Total Orders

Shows the number of completed customer orders.

Business Insight

Order volume helps evaluate overall business activity and customer engagement.

KPI 3 – Average Order Value (AOV)

Average revenue generated from each order.

Business Insight

A higher AOV suggests customers purchase more products per transaction.

KPI 4 – Average Shipping Time

Measures the average number of days required to deliver customer orders.

Business Insight

Lower shipping time generally improves customer satisfaction.

KPI 5 – Unique Customers

Represents the total number of customers served.

Business Insight

This metric helps evaluate customer reach and business growth.

KPI 6 – Revenue per Customer

Measures the average revenue generated by each customer.

Business Insight

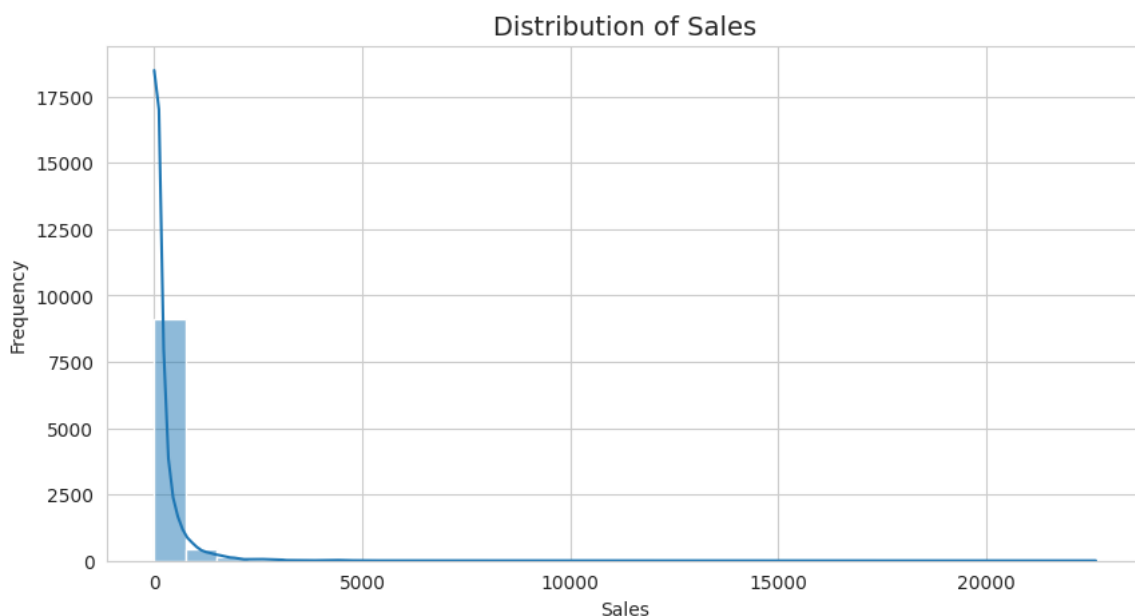
Useful for evaluating customer lifetime value and marketing effectiveness.

8. Exploratory Data Analysis

Sales Distribution

The sales distribution shows that most transactions involve relatively small purchase amounts, while only a limited number of transactions generate exceptionally high sales. This indicates a right-skewed distribution, which is common in retail businesses.

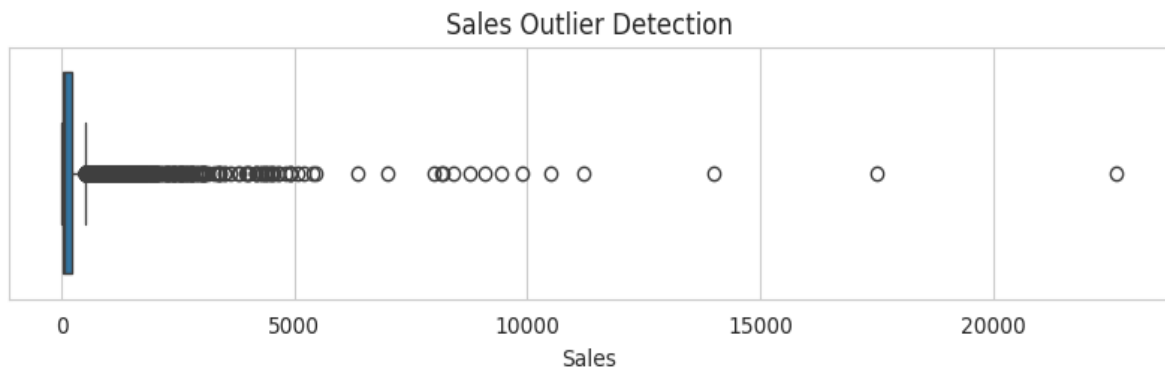
Figure 1.



Sales Outlier Detection

Several high-value transactions were identified as outliers. These large orders significantly contribute to overall revenue and should be analyzed separately to understand enterprise or bulk customer purchasing behavior.

Figure 2.

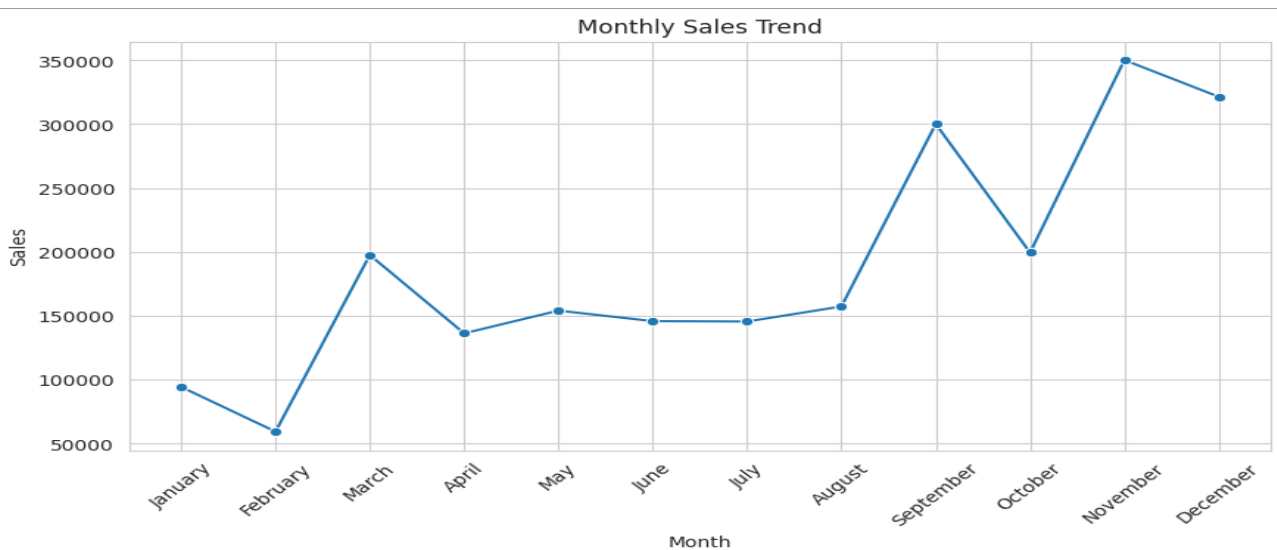


9. Business Analysis

Monthly Sales Trend

The monthly sales trend demonstrates fluctuations throughout the year. Certain months consistently generate higher sales, indicating seasonal purchasing behavior. Businesses should prepare inventory and promotional campaigns during these peak periods.

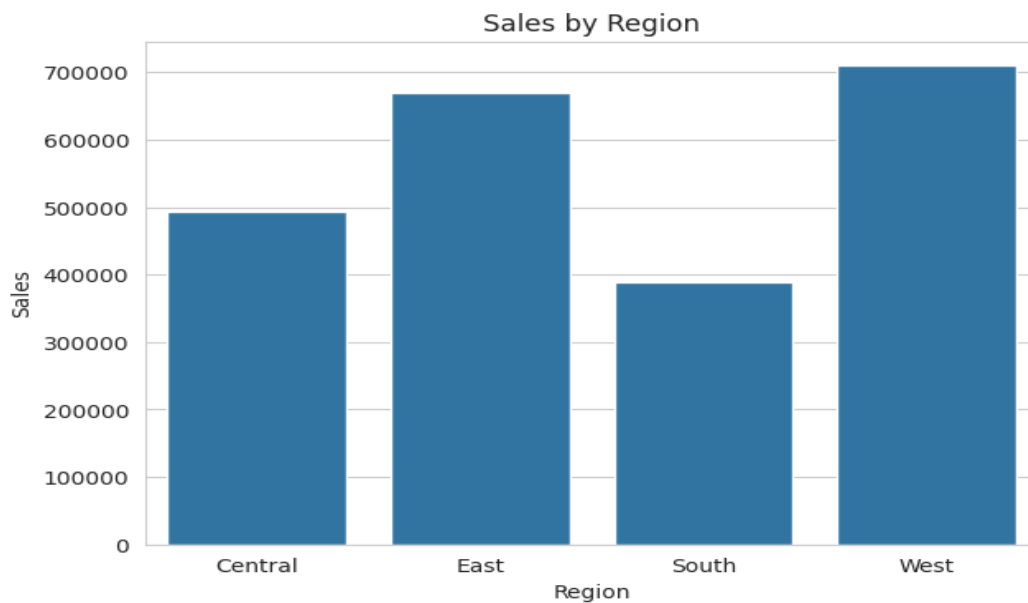
Figure 3.



Sales by Region

Regional analysis highlights differences in revenue contribution among geographical locations. High-performing regions represent strong market opportunities, while low-performing regions require targeted marketing strategies and customer engagement.

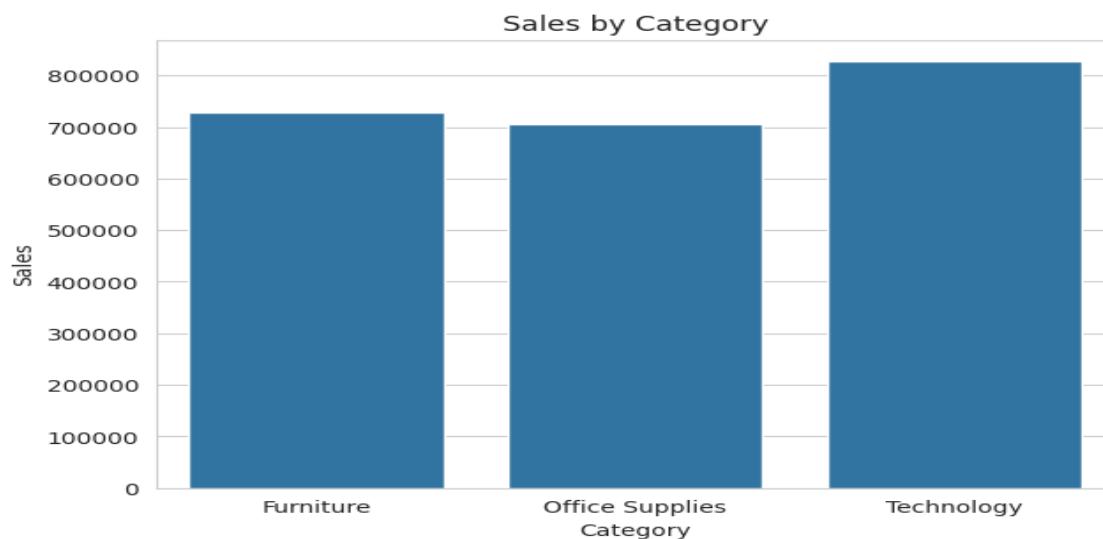
Figure 4.



Sales by Category

Product categories contribute differently to total revenue. Categories generating higher sales indicate stronger customer demand, while weaker categories may require pricing adjustments, promotions, or product improvements.

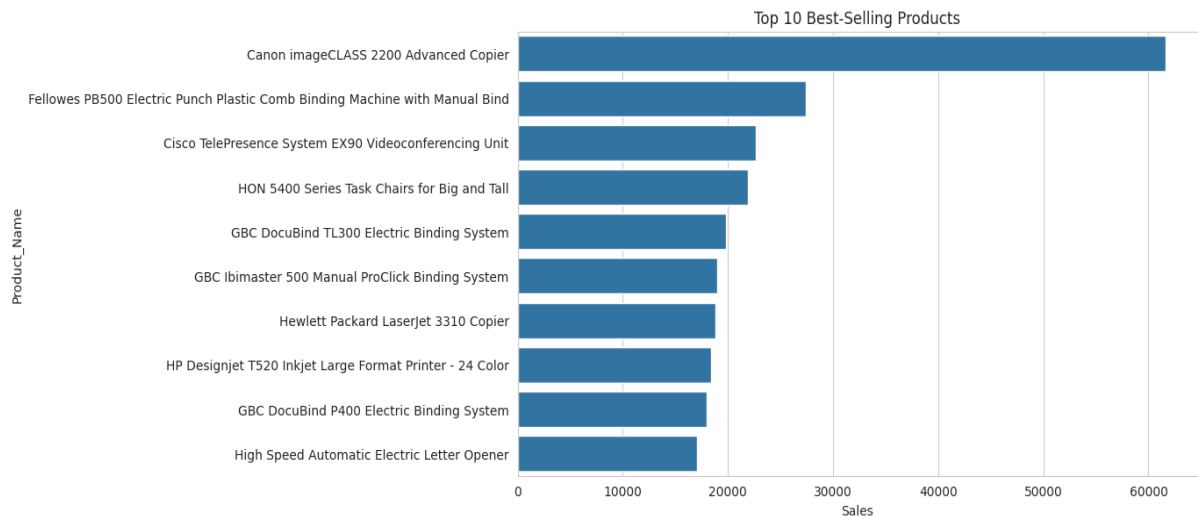
Figure 5.



Top 10 Products

The top-selling products contribute significantly to overall business revenue. These products should remain well-stocked and receive priority in promotional campaigns to maximize sales.

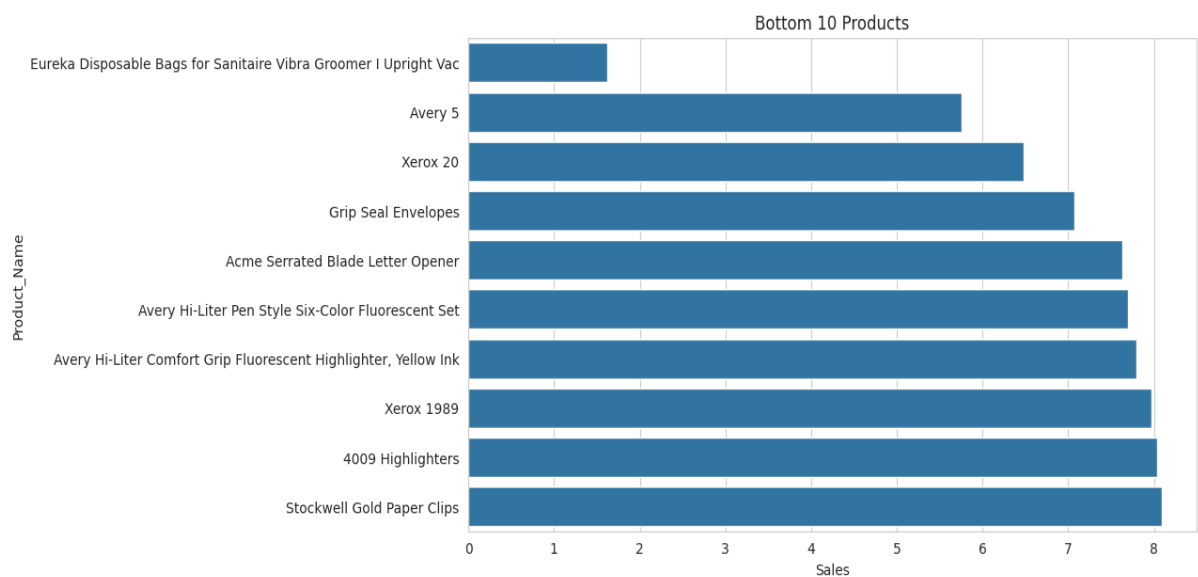
Figure 6.



Bottom 10 Products

Low-performing products generate minimal sales and may increase inventory holding costs. Businesses should evaluate whether these products need repositioning, discounts, or discontinuation.

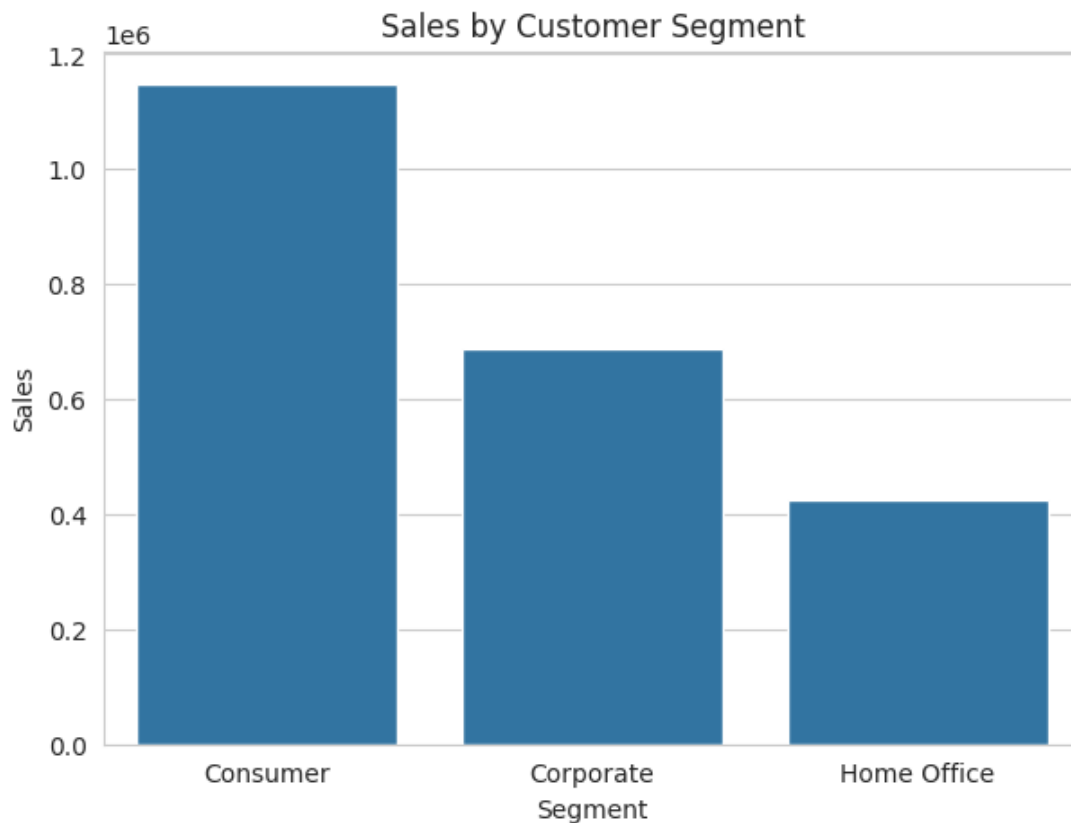
Figure 7.



Sales by Customer Segment

Different customer segments contribute differently to business revenue. Identifying high-value customer groups enables personalized marketing and improved customer retention strategies.

Figure 8.



10. Overall Findings

The analysis revealed several important business insights:

- Sales are not evenly distributed throughout the year.
- Some regions consistently outperform others.
- Product categories contribute differently to revenue.
- A small number of products generate a large portion of total sales.
- Several products underperform and require strategic review.
- Customer purchasing behavior varies across segments.
- Shipping performance remains an important factor influencing customer satisfaction.

11. Recommendations

Based on the analysis, the following recommendations are proposed for Alfido Tech:

1. Focus on High-Performing Products

Increase inventory and promotional efforts for top-selling products to maximize revenue.

2. Improve Low-Performing Products

Review pricing strategies, product quality, and promotional campaigns for products with consistently low sales.

3. Expand Successful Regional Strategies

Identify successful marketing approaches used in high-performing regions and implement similar strategies in weaker regions.

4. Prepare for Seasonal Demand

Use historical sales trends to forecast inventory requirements and launch seasonal marketing campaigns during peak sales months.

5. Increase Average Order Value

Introduce bundle offers, product recommendations, and cross-selling strategies to encourage customers to purchase additional products.

12. Business Impact

The analysis helps organizations:

- Improve revenue forecasting
- Optimize inventory management
- Identify profitable products
- Improve customer satisfaction
- Support data-driven business decisions
- Increase operational efficiency

13. Conclusion

This project successfully analyzed sales performance using the Superstore Sales dataset through comprehensive data preprocessing, exploratory analysis, KPI measurement, and business visualization. The findings provide valuable insights into customer behavior, product performance, regional sales distribution, and seasonal trends.

The recommendations presented can help organizations like Alfido Tech improve sales performance, optimize resource allocation, and develop more effective business strategies. This project demonstrates how data analytics can transform raw business data into meaningful insights that support informed decision-making.

14. References

- Kaggle Superstore Sales Dataset
<https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>
- Python Documentation
<https://docs.python.org/3/>
- Pandas Documentation
<https://pandas.pydata.org/>
- NumPy Documentation
<https://numpy.org/>